

Requirements Table

Second-Hand Product Escrow Marketplace — Problem Statement #39

Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall lock the buyer's payment in escrow and set the order status to 'Under Inspection' until the Inspection Agent submits a sign-off.	High	Pass: funds are held and released only after dual verification (inspection + buyer approval). Fail: funds reach the seller before both approvals are recorded.	Protects both parties until the product's condition is verified.
FR-002	Functional	The system shall allow a Buyer to raise a dispute on an order at any point before escrow funds are released.	High	Pass: submitting a dispute freezes the order's escrow funds within 5 seconds and creates a ticket visible to the Admin. Fail: funds can still be released after a dispute is filed.	Gives buyers recourse when the delivered product does not match the inspection report.
FR-003	Functional	The system shall allow a Seller to list a product with a title, price, description, and at least one photo.	High	Pass: a listing missing any required field is rejected with a specific error message. Fail: an incomplete listing is published to the marketplace.	Ensures buyers have enough information to make a purchase decision.
FR-004	Functional	The system shall allow an Inspection Agent to submit a pass/fail inspection report with notes for an order assigned to them.	Medium	Pass: a submitted report changes the order state and is timestamped and visible to buyer, seller, and admin. Fail: an order proceeds to buyer sign-off without a report on file.	Independent verification is the core trust mechanism of the marketplace.
FR-005	Functional	The system shall notify the Buyer and Seller by email/in-app alert whenever an order's status changes (inspection complete, dispute raised, funds released).	Medium	Pass: a status change triggers a notification to both parties within 1 minute. Fail: a status change occurs with no notification sent.	Keeps both parties informed without requiring manual status checks.
NFR-001	Nonfunctional	Dispute resolution tickets shall freeze escrow funds immediately and block any withdrawal attempts while the dispute is under investigation.	High	Pass: benchmarking confirms the freeze applies within the target latency and blocks withdrawal attempts under simulated peak load. Fail: a withdrawal succeeds during an active dispute.	Prevents funds from being moved out of reach while a dispute is being investigated.
NFR-002	Nonfunctional	The system shall encrypt all payment and escrow transaction data in transit and at rest using industry-standard encryption.	High	Pass: a security audit confirms TLS is enforced in transit and AES-256 (or equivalent) at rest, with no plaintext financial data found. Fail: any financial data is found unencrypted.	Protects sensitive financial data given the platform's escrow/payment responsibilities.